

# Topology-guided Perception-aware Receding Horizon Trajectory Generation for UAVs

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**Abstract**—The perception-aware motion planning method based on the localization uncertainty has the potential to improve the localization accuracy for robot navigation. However, most of the existing perception-aware methods pre-build a global feature map and can not generate the perception-aware trajectory in real time. This paper proposes a topology-guided perception-aware receding horizon trajectory generation method, which contains a topology-guided position trajectory generation and a perception-aware yaw angle trajectory generation. Specifically, a memorable active map is built by selectively storing the visual landmarks. After that, a library of candidate topological trajectories are generated, which are then evaluated in terms of the perception quality based on the active map, smoothness, collision possibility and feasibility. In addition, the yaw angle trajectory is obtained through a front-end multiple refined path search and a back-end path-guided trajectory optimization. Comparative simulation and real-world experiments are carried out to confirm that the proposed method can keep more visual features in view and reduce the localization error.

## I. INTRODUCTION

Unmanned aerial vehicles (UAVs) are widely used for different tasks such as autonomous exploration [1], mapping [2], photography [3], goods transportation [4] and rescue [5]. In these applications, the state estimation module is crucial to obtain the position, velocity and rotation of UAVs. The camera and inertial measurement unit (IMU) are appropriate sensors for the state estimation of UAVs due to their low cost and weight. Typical visual-inertial navigation systems track multiple visual features to estimate the state of the robot. The motion planning module will affect the observation of visual features. Meanwhile, the quantity and the quality of the visual features in the view will also affect the localization accuracy. However, most existing motion planning methods focus on the time-optimal performance and the safety of

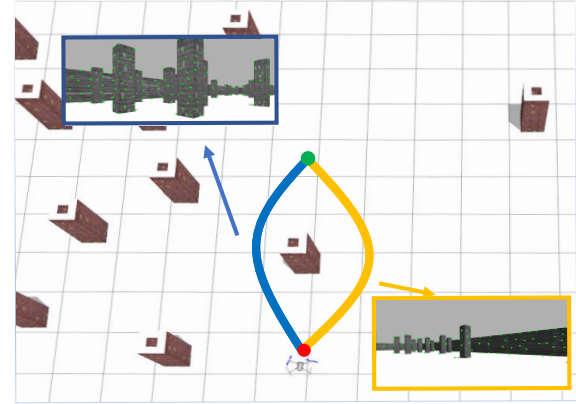


Fig. 1. Illustration of different motion planning methods for a quadrotor from the red starting point to the green ending point. The blue trajectory with perception awareness enables the quadrotor to see the regions with richer textures. On the other hand, the yellow trajectory, lacking perception awareness, is susceptible to crossing the areas with the fewer textures, resulting in inaccurate localization.

the position trajectory [6]–[11], which do not consider the perception constraint.

It is called *perception-aware planning* to consider the perception quality in motion planning. As shown in Fig. 1, during the flight, the trajectory generation method considering the perception quality will choose the path with more visual features as well as face the feature-rich areas, obtaining the higher localization accuracy. On the contrary, perception-agnostic motion planning will potentially result in facing featureless areas and then reducing the localization accuracy. Therefore, motion planning should consider the perception quality. Some studies focus on the perception-aware planning, while there are still some open problems listed as follows:

- 1) Most of the existing perception-aware planning methods pre-build the global feature map, reusing the visual features to generate a perception-aware trajectory. However, the trajectory can not be calculated in real time.
- 2) According to the differential flatness, the quadrotor motion planning problem can be decoupled into the position and yaw angle planning. However, the yaw perception-aware trajectory is often ignored.

## A. Contributions

To address the issues above, we propose a topology-guided receding horizon method for generating perception-aware trajectories for UAVs. First, we build a memorable active map that stores the visual features from the current

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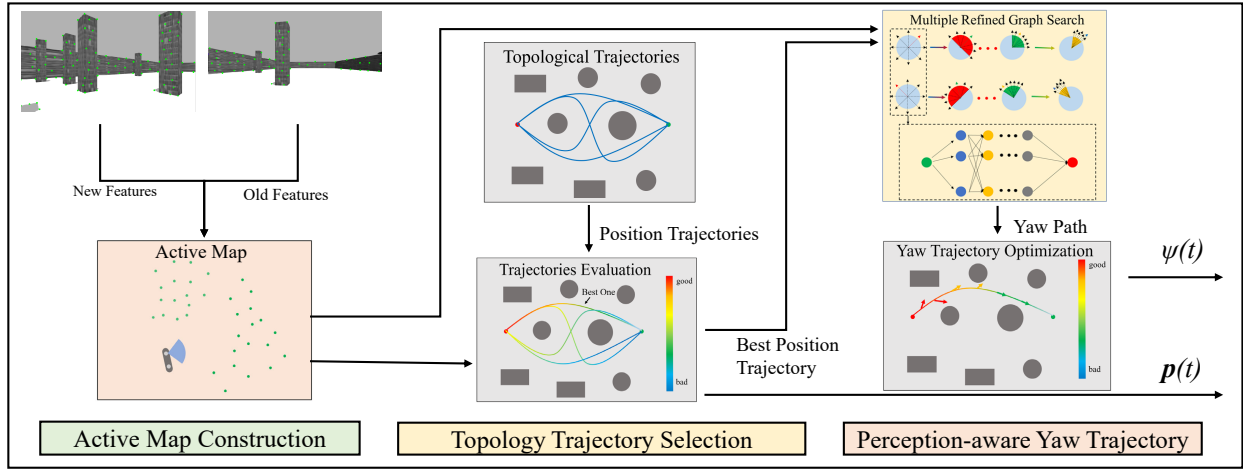


Fig. 2. System Overview

and historical frames. Then, we generate a series of topological trajectories using an ESDF-free (Euclidean Signed Distance Field) trajectory generation method incorporating the proposed active map. The best trajectory is selected by evaluating the perception quality, collision probability, smoothness, and dynamic feasibility. Next, we utilize a two-stage trajectory generation method to obtain the yaw trajectory. In the first stage, a multiple refined path search method is proposed to find a perception-aware and safe path, and an initial yaw trajectory is generated from this path. In the second stage, a path-guided trajectory generation considers the smoothness and dynamic feasibility. The position and yaw trajectories are all parameterized to the uniform B-splines. The convex hull property of the B-spline is utilized to ensure safety and dynamic feasibility. The contributions of this paper are summarized as follows:

- 1) A memorable active map generation method is put forward to evaluate the perception information in real time. The map takes the advantage of the historical features to improve the perception quality.
- 2) We propose a perception-aware receding horizon trajectory generation method, which includes a topology-guided position trajectory and a perception-aware yaw angle trajectory. The trajectory can actively observe more features and reduce the localization uncertainty.
- 3) Comparative simulation and real-world tests are conducted to verify the superior performance of the localization accuracy.

## II. RELATED WORK

Perception-aware planning for UAVs has been gradually concerned in recent years. Zhou *et al.* [12] propose a robust and perception-aware trajectory replanning system, which aims to observe the unknown environment actively. Jesus *et al.* [13] propose a real-time optimization-based perception-aware trajectory planner for UAVs in dynamic environments, which aims to avoid dynamic obstacles and keep them in the sensor field of view. After that, Jesus *et al.* [14] further propose a learning-based perception-aware trajectory planner, which can achieve a similar cost and a lower time cost

compared with the optimization-based method. However, the methods mentioned above aim to avoid dynamic obstacles and actively observing the unknown environment instead of reducing the localization uncertainty.

Some perception-aware planning methods aim to reduce the localization uncertainty. Varun *et al.* [15] propose a method by maximizing the co-visibility of the features to increase the estimation accuracy. However, the method lacks the obstacle avoidance. Sabdat *et al.* [16] propose a perception-aware planning method, which considers the perception quality in a sampling-based Rapidly-exploring Random Tree (RRT) for motion planning. Sampling-based planners [17] may suffer from sampling inefficiency, which results in much computational cost to find a global path. Therefore, Dabin *et al.* [18] propose a topology-guided path planning strategy for perception-aware navigation. Zhang *et al.* [19] propose to utilize fisher information to evaluate the perception quality and build a Fisher Information Field (FIF) for the trajectory planning. However, the methods mentioned above all pre-build a global map, which cannot generate the trajectory in real time.

Some recent methods focus on generating perception-aware trajectory in a receding horizon fashion. Reiya *et al.* [20] propose an informative path planning framework with a lidar-based state estimation system, while it is designed based on a lidar model. Zhang *et al.* [21] use the information from the monocular state estimation and mapping system to evaluate a library of candidate trajectories generated by a motion-primitive method. Wu *et al.* [22] propose a receding horizon planner, which can adjust the trajectory aggressiveness and reduce the motion blur. However, these two methods do not plan the yaw trajectory to minimize the localization uncertainty. Motivated by the observations, this paper proposes a topology-guided perception-aware receding horizon trajectory generation method, which contains a topology-guided position trajectory generation for fully exploring the environment and a perception-aware yaw angle generation for minimizing the localization uncertainty.

The rest of this paper is structured as follows. In Section III, the overview of the topology-guided perception-aware

receding horizon system is given. In Section IV, we detail the process of building a memorable active map and how to incorporate the active map to generate perception-aware trajectories. Section V introduces a perception-aware yaw angle trajectory generation process. To demonstrate the performance of the proposed method, simulation and real-world experiments are shown in Section VI.

### III. SYSTEM OVERVIEW

The system overview is shown in Fig. 2, consisting of three modules: active map construction, topology trajectory selection and perception-aware yaw trajectory generation. Firstly, a memorable active map is constructed using odometry [23] and features information from the current frame and historical frames. Secondly, multiple topological trajectories are generated based on the grid map, and the optimal trajectory is selected by evaluating the perception information, collision likelihood, smoothness, and dynamic feasibility. After that, a yaw angle trajectory is generated for the optimal position trajectory. Specifically, the front end is a multiple refined graph search strategy to find a yaw path with perception awareness, and then optimizing the yaw trajectory by adding constraints including smoothness, feasibility, and safety. Finally, a complete perception-aware trajectory is obtained.

### IV. TOPOLOGY-GUIDED POSITION TRAJECTORY GENERATION

#### A. Active Map and Perception Quality Metric

A memorable active map is constructed to evaluate the perception quality as shown in Fig. 3. Specifically, VINS-fusion outputs a set of sparse features observed and tracked from different frames, which are also called *Landmarks*. Different from [21], the active map in this method stores the features not only in the current frame but also in the history frame. Due to the limited observation region, the quadrotor may miss the texture-rich areas when replanning the trajectory, resulting in poor perception quality. Therefore, we maintain a fixed-size priority queue to store the landmarks. When a new frame arrives, all the landmarks in the current frame enter the queue, and the landmarks in the last frame that overlaps with the current frame are dequeued. The remaining older landmarks are dequeued based on the current queue size. Let  $M_A$  represent the active map with  $K$  features, which can be mathematically expressed as

$$M_A = \{\mathbf{l}_i\}_{i=1}^K, \quad (1)$$

$$K = k_{\text{older}} + k_{\text{last}} + k_{\text{cur}}, \quad (2)$$

where  $\mathbf{l}_i \in \mathbb{R}^3$  stands for the  $i$ -th feature in the active map.  $k_{\text{older}}$ ,  $k_{\text{last}}$  and  $k_{\text{cur}}$  represent the feature numbers of the older frame, last frame and the current frame, respectively. Let  $O_k$  denote the landmarks set observed at pose  $T_k \in SE(3)$ . After generating the active map, a least square problem is constructed to estimate the pose, which is formulated as

$$T_k^* = \arg \min_T \sum_{j \in O_k} \|\tilde{\mathbf{u}}_{kj} - \text{proj}(T_c^b T^{-1} \mathbf{l}_j)\|^2 \quad (3)$$

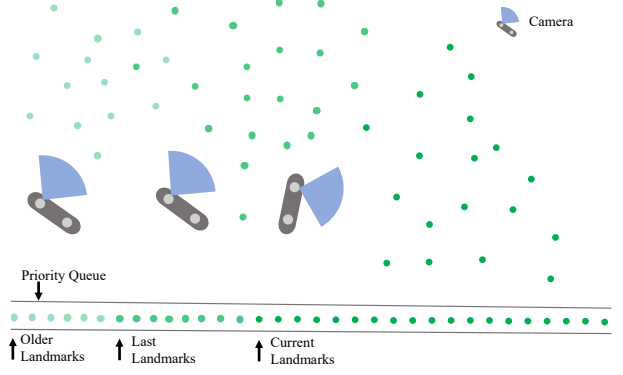


Fig. 3. The memorable active map generated by the output of VINS-fusion. The fixed-size priority queue stores the landmarks including the current frame, last frame and older frame.

where  $\text{proj}(\cdot)$  is the projection function of the camera.  $T_c^b \in SE(3)$  is the transformation matrix from the camera frame to the body frame and  $\tilde{\mathbf{u}}_{kj} \in \mathbb{R}^{2 \times 1}$  is the noisy observation of the  $j$ -th landmark. As a nonlinear optimization problem, it can be solved using the iterative optimization method.  $T_k$  is updated as  $T_k \leftarrow T_k \text{Exp}(\xi)$ , where  $\text{Exp}(\cdot)$  is the map from  $\xi \in \mathfrak{se}(3)$  to the element in  $SE(3)$ . The problem is solved by linearizing around the current estimate of  $T_k$  as

$$\xi^* = \arg \min_{\xi} \sum_{j \in O_k} \|\mathbf{u}_{kj} - \tilde{\mathbf{u}}_{kj} + J_{\xi}^{kj} \xi\|^2, \quad (4)$$

where  $\mathbf{u}_{kj} = \text{proj}(T_c^b T_k^{-1} \mathbf{l}_j)$  and  $J_{\xi}^{kj} \in \mathbb{R}^{2 \times 6} = \frac{\partial \mathbf{u}_{kj}}{\partial \xi}$ . Then the information matrix can be obtained as

$$H_k = J_k^T \Sigma_u^{-1} J_k \quad (5)$$

where  $J_k$  is stacked up by  $J_{\xi}^{kj}$ .  $\Sigma_u$  is the covariance matrix of  $\{\mathbf{u}_{kj}\}_{j \in O_k}$ , which is usually with the same value on the diagonal when the observations are independent and have the same noise level in a visual inertial state estimation. The whole information matrix is formulated as

$$H = \begin{bmatrix} H_1 & 0 & \dots & 0 \\ 0 & H_2 & \dots & 0 \\ 0 & \dots & \dots & 0 \\ 0 & \dots & 0 & H_K \end{bmatrix} \quad (6)$$

Utilizing (6), the perception information cost using an analog form of the D-opt criterion following [19] is:

$$c_{\text{perc}} = -\exp\left(\frac{1}{6K} \log(\det(H))\right) \quad (7)$$

#### B. Topology-guided trajectory

Homotopy is a widely used concept in motion planning to generate topological trajectories, while it captures insufficient useful trajectories. As defined in [24], we use uniform visibility deformation (UVD) to check whether the two trajectories are topologically distinctive. As shown in Fig. 4, four topologically distinctive trajectories are generated from the start to the end. For each trajectory, the deeper the color, the higher the perception quality. The best trajectory is chosen by evaluating the perception awareness, collision

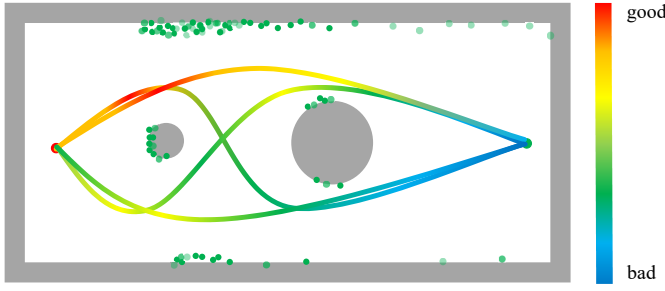


Fig. 4. Illustration of four topologically distinctive trajectories. The trajectories are generated by the current memorable active map. The green points stand for the landmarks stored by the active map, wherein the different depth indicates the quantity of the perception information.

probability, smoothness and dynamic feasibility. The specific evaluation function is designed as

$$\text{cost} = k_{\text{per}}c_{\text{per}} + k_sc_s + k_cc_c + k_dc_d \quad (8)$$

where  $c_{\text{per}}$ ,  $c_s$ ,  $c_c$  and  $c_d \in \mathbb{R}$  stand for the perception information cost, smoothness cost, collision cost and dynamic feasibility cost, respectively.  $k_{\text{per}}$ ,  $k_s$ ,  $k_c$  and  $k_d \in \mathbb{R}^+$  are positive weights for each penalty term. After the optimization, a final position trajectory is obtained, which is perception-aware, smooth, safe and feasible.

## V. PERCEPTION-AWARE YAW TRAJECTORY

### A. Multiple Refined Local Path Search

In most previous motion planning methods, the yaw angle is either not planned or along the direction of the projection of the trajectory tangent direction in the  $x$ - $y$  plane, lacking the perception awareness. We plan a perception-aware yaw trajectory to actively observe more textured areas during flight, thus reducing localization uncertainty. The position trajectory is sampled into multiple points at uniform time intervals. For each point, multiple yaw angles are uniformly sampled from  $-\pi$  to  $\pi$ . Each position  $\mathbf{p} \in \mathbb{R}^3$  and each angle  $\psi \in \mathbb{R}$  form a pair as a search node and different angles at the same position form a layer. Next, we construct a graph connecting nodes in consecutive layers through directed edges. The nodes in the same layer share the same time and position values. To obtain a perception-aware, safe and smooth trajectory, the cost function is designed as

$$c = k_p c_{\text{per}} + k_\psi c_\psi + k_{\Psi_{\text{max}}} c_{\Psi_{\text{max}}} + k_{\text{saf}} c_{\text{saf}}, \quad (9)$$

where  $k_p$ ,  $k_\psi$ ,  $k_{\Psi_{\text{max}}}$ , and  $k_{\text{saf}} \in \mathbb{R}$  are positive weights. The cost term for perception awareness  $c_{\text{per}}$  is the same as the one designed for the position trajectory. The second cost term is designed as

$$c_\psi = (\text{wrap}_{-\pi}^\pi(\psi_{n_2} - \psi_{n_1}))^2 \quad (10)$$

where  $\text{wrap}_{-\pi}^\pi(\cdot)$  means wrapping of an angle in  $[-\pi, \pi)$  and  $c_\psi \in \mathbb{R}$  is a smooth cost to penalize the change between two near angles. The cost term  $c_{\Psi_{\text{max}}} \in \mathbb{R}$  is designed as

$$c_{\Psi_{\text{max}}} = \mathbf{1}_{|\text{wrap}_{-\pi}^\pi(\psi_{n_2} - \psi_{n_1})| / (t_{n_2} - t_{n_1}) > \Psi_{\text{max}}} \quad (11)$$

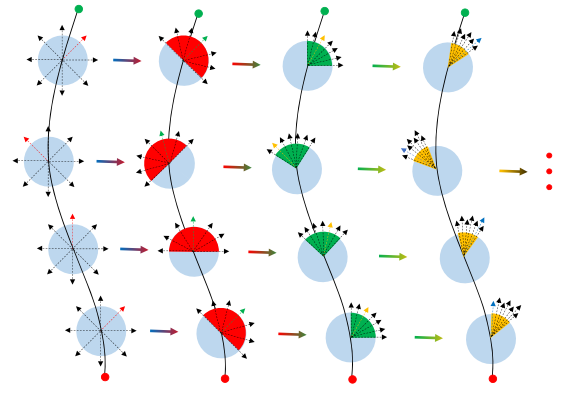


Fig. 5. Illustration of the multiple refined graph search. The arrows represent the sampled yaw angles, wherein the colored arrows represent the search results. The sampled area is halved after each search, which is represented by the colored sector.

where  $\Psi_{\text{max}} \in \mathbb{R}^+$  is the maximum yaw velocity and  $c_{\Psi_{\text{max}}}$  is designed for satisfying the dynamic feasibility and reducing the motion blur. When the current yaw angle velocity exceeds the maximum yaw angle velocity, the cost is 1, otherwise, the cost is 0. Due to the limited field of view (FOV) of cameras, the quadrotor is unable to observe all-around obstacles. For ensuring the flight safety, a safety cost is designed as

$$c_{\text{saf}} = (\psi - \psi_{\text{tag}})^2 \quad (12)$$

where  $\psi_{\text{tag}} \in [-\pi, \pi)$  is along the tangent direction of the position trajectory. The cost is designed to keep the quadrotor not deviating too much from the forward direction, avoiding losing the visibility of the obstacles in the front. The root node and the goal node are the yaw angles of the start and end positions, respectively.

The problem is solved by *Dijkstra* algorithm. As illustrated in Fig. 5, a multiple refined path search strategy is put forward to search the path more efficiently, avoiding abundant useless nodes. To search for a fine trajectory, the sampling interval is usually very small, resulting in a large number of useless search nodes. Therefore, we first sample at a larger interval and gradually narrow the sampling area based on the path searched. The search process ends when the costs of two searches do not differ much. Meanwhile, the planning is based on current sensor data for receding horizon planning. However, during the execution, obstacles may suddenly be encountered, requiring replanning. If the yaw angle is not facing forward at this time, a collision may occur. Thus, the yaw angle search is limited to a certain horizon, not searching the entire trajectory. The latter part is along the tangential direction of the trajectory.

### B. Path-guided Trajectory Generation

The initial B-spline trajectory is generated guided by the searched path with the same time allocation as the position trajectory. The trajectory is represented as B-spline curve

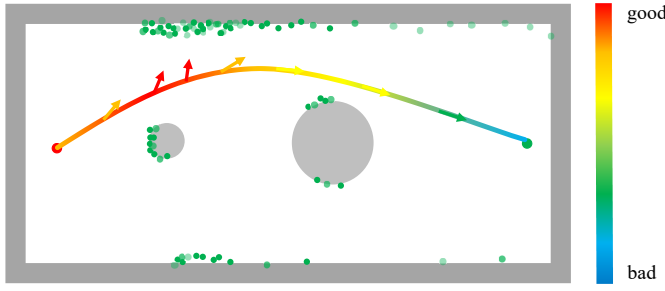


Fig. 6. The trajectory after generating the yaw angle trajectory for the best position trajectory. The arrows represent the yaw angle direction samplings and the colors indicate the perception quality.

with  $N + 1$  control points  $\{Q_0, Q_1, \dots, Q_N\}$ . The initial trajectory guides the trajectory to ensure perception awareness, which is also sampled uniformly with  $N + 1$  control points  $\{B_0, B_1, \dots, B_N\}$ . The convex hull property of the B-spline is employed to limit the yaw angular velocity and acceleration. Meanwhile, the smoothness of the trajectory is also considered. A nonlinear optimization problem is constructed as follows:

$$\begin{aligned} \min \quad & \alpha_i \sum_{i=0}^N (Q_i - B_i)^2 + \alpha_s \sum_{i=1}^{N-1} (Q_{i+1} - 2Q_i + Q_{i-1})^2 + \\ & \alpha_v \sum_{i=1}^{N-1} (V_i^2 - V_{\max}^2)^2 \cdot \mathbf{1}_{|V_i| \geq V_{\max}} + \\ & \alpha_a \sum_{i=2}^{N-2} (A_i^2 - A_{\max}^2)^2 \cdot \mathbf{1}_{|A_i| > A_{\max}} \end{aligned}$$

The first term penalizes the error between the current trajectory and the guided path to increase the perception awareness. The second term ensures the smoothness of the trajectory. The third term limits the speed of the trajectory to be less than the maximum speed, where  $V_i \in \mathbb{R}$  represents the velocity control point and  $V_{\max} \in \mathbb{R}^+$  represents the maximum velocity. The fourth term restricts the acceleration of the trajectory to be less than the maximum acceleration, where  $A_i \in \mathbb{R}$  represents the acceleration control point and  $A_{\max} \in \mathbb{R}^+$  represents the maximum acceleration. And  $\alpha_i, \alpha_s, \alpha_v$  and  $\alpha_a \in \mathbb{R}^+$  are positive weights. The position, velocity, and acceleration constraints of the starting and ending points are added as hard constraints to the optimization problem. A safe, smooth, perception-aware and dynamically feasible trajectory is acquired after the optimization as shown in Fig. 6.

## VI. RESULTS

Both the comparative simulation and real-world experiments are conducted to verify the superior performance of the proposed method. Specifically, we compare the number of features, end position error and root mean square error (RMSE) between perception-agnostic ego-planner [25] and the proposed method.

### A. Perception Awareness

The simulation is conducted using the XTDrone platform [26]. Two scenarios are designed to verify the performance of

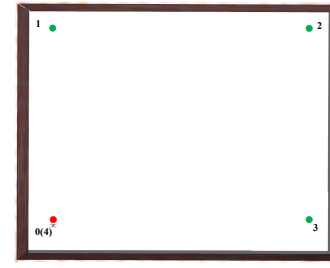


Fig. 7. Illustration of *obstacle-free* simulation scene.

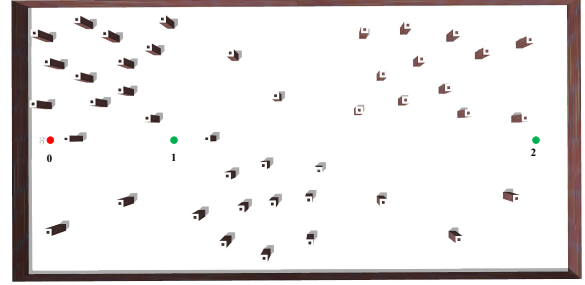


Fig. 8. Illustration of *obstacle* simulation scene.

the method. Under the *obstacle-free* environment, as shown in Fig. 7, the quadrotor rotates around the scene once and returns to the starting position. Since there are only walls on four sides and the target location is inside the walls, it can be approximated as an unobstructed environment. The second scenario is the *obstacle* environment, as shown in Fig. 8. In this scenario, there are many cylindrical obstacles, and the textures on the obstacles are the same.

The state estimation is prone to accumulating drift over time, decreasing the localization accuracy. In this regard, the perception-agnostic method is more likely to result in localization inaccuracy as the quadrotor may fly in areas with fewer textures. For instance, Fig. 9 shows the results of the perception-aware method and ego-planner in an *obstacle-free* environment. It is evident from Fig. 9(a) that the odometry of the perception-aware method is closer to the ground truth. Fig. 10 shows the comparative results in an *obstacle* environment. Considering perception awareness in motion planning allows the quadrotor to fly facing areas with richer textures, which is also beneficial for feature extraction by visual-inertial odometry. The number of features extracted by the left camera in each frame is also recorded to reflect the perception awareness, as shown in Fig. 11. The perception-aware method can extract more visual features. The quantitative results of feature count in different scenarios are listed in Tabs. I-II, respectively. In the *obstacle-free* scenario, the mean number of features can be increased by 10.76%. In the *obstacle* scenario, it can be increased by 20.77%.

### B. Localization Accuracy

To reduce the localization uncertainty and reach the destination more accurately while ensuring safety, a topology-guided receding horizon perception-aware planning strategy is proposed in this paper. As shown in Fig. 12, the quadrotor generates two topologically distinctive trajectories during the

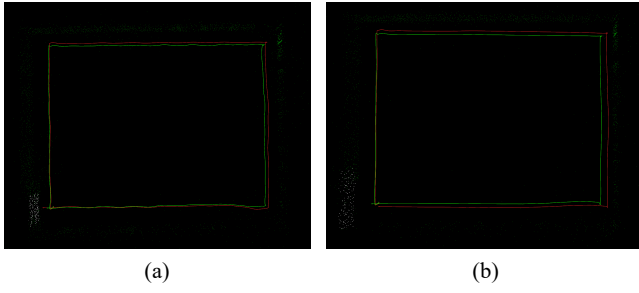


Fig. 9. The trajectories generation under *obstacle-free* scene in a receding horizon fashion. The red line represents the ground truth trajectory, the green line represents the odometry output. (a): proposed method (b): ego-planner

TABLE I  
COMPARISON OF DIFFERENT INDICATORS IN *obstacle-free* SCENE

| indicator              | ego-planner | proposed     | difference |
|------------------------|-------------|--------------|------------|
| mean feature number    | 86.58       | <b>95.9</b>  | +10.76%    |
| end position error (m) | 0.306       | <b>0.157</b> | -48.69%    |
| RMSE (m)               | 0.214       | <b>0.123</b> | -42.52%    |

TABLE II  
COMPARISON OF DIFFERENT INDICATORS IN *obstacle* SCENE

| indicator              | ego-planner | proposed      | difference |
|------------------------|-------------|---------------|------------|
| mean feature number    | 83.69       | <b>101.07</b> | +20.77%    |
| end position error (m) | 0.782       | <b>0.405</b>  | -48.2%     |
| RMSE (m)               | 0.412       | <b>0.25</b>   | -39.32%    |

flight. After evaluating the perception quality, collision possibility, smoothness, and dynamic feasibility of the trajectories, the optimal trajectory, i.e., the red trajectory, is selected for execution. Meanwhile, the yaw angle planning is performed during flight to face areas with richer textures, reducing the localization uncertainty. To quantitatively evaluate the localization quality, we compare the position errors of the two methods, including end position error and RMSE, in two different scenarios. As listed in Tabs. I-II, the perception-aware method can effectively reduce errors compared with the perception-agnostic method. In the *obstacle-free* scenario, the end position error can be reduced by 48.69%, and RMSE can be reduced by 42.52%. In the *obstacle* scenario, the end position error and RMSE can be reduced by 48.2% and 39.2%, respectively. The average computation time of the proposed method is 2.25ms and the real time performance is promised.

### C. Real-World Experiments

A homemade quadrotor shown in Fig. 13(a) is employed to validate the effectiveness of the method, which is equipped with an Intel mini-computer to run the algorithms and controlled by PIXHAWK mini. VINS-fusion is employed to estimate the state, wherein the image input is provided by an Intel Realsense D435i camera. In addition, it can output the features to construct a memorable active map. The depth measurement provided by D435i is used to provide point clouds for obstacle avoidance. Moreover, the T265 camera

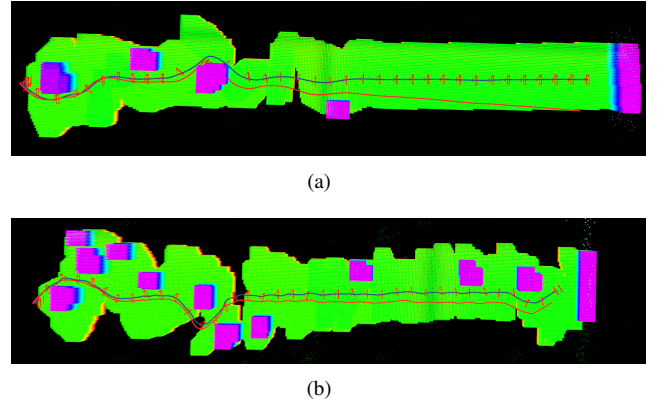


Fig. 10. The trajectories generation under *obstacle* scene in a receding horizon fashion. (a): The trajectory planned by ego-planner. (b): The trajectory generated by the proposed method. The red line represents the ground truth trajectory, and the blue line represents the odometry output. The red markers indicate the heading direction of the quadrotor.

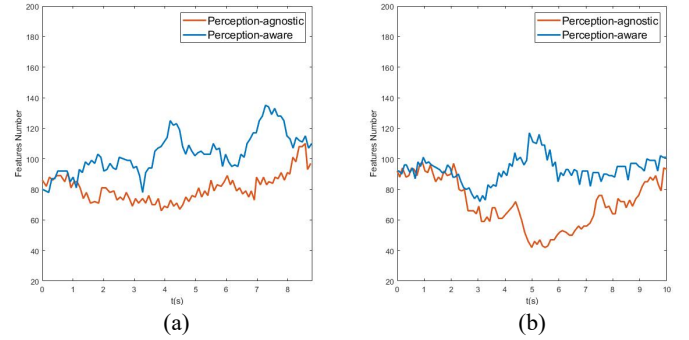


Fig. 11. The variation of the feature number over time under two scenes *obstacle* and *obstacle-free*. (a): The number variation of two methods from point 0 to point 1 under the scene *obstacle-free*. (b): The number variation of topological distinctive trajectories from point 0 to point 1 under scene *obstacle*.

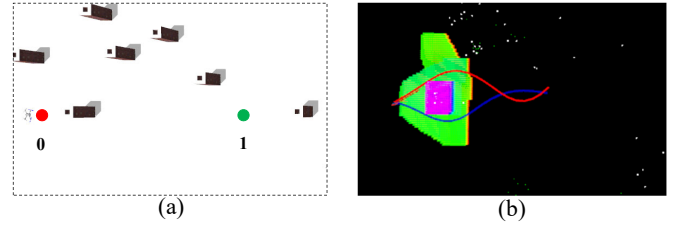


Fig. 12. Topologically distinctive trajectories.

is utilized to provide the ground truth.

We test the proposed method in a real-world scene, which is shown in Fig. 13(b). During the flight, the maximum speed is set to 1m/s, and other parameters are same as in the simulation. Similar to the simulation results, our system was able to command the quadrotor to follow a perception-aware trajectory to reach the goal. Fig. 14 shows the error mapped onto the trajectory on the  $x$ -axis and  $y$ -axis.

## VII. CONCLUSION

This paper studies taking perception quality into the motion planning problem to improve the localization accuracy for the quadrotors. Unlike most previous methods that pre-build a global map, the proposed method plans the trajectory

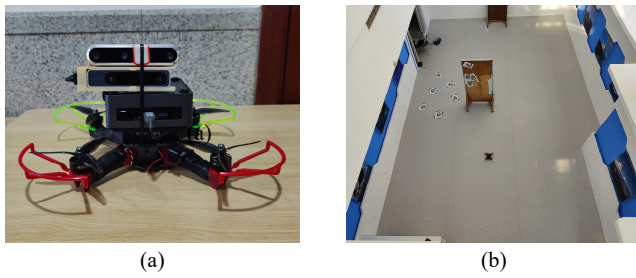


Fig. 13. The quadrotor platform and experiment field.

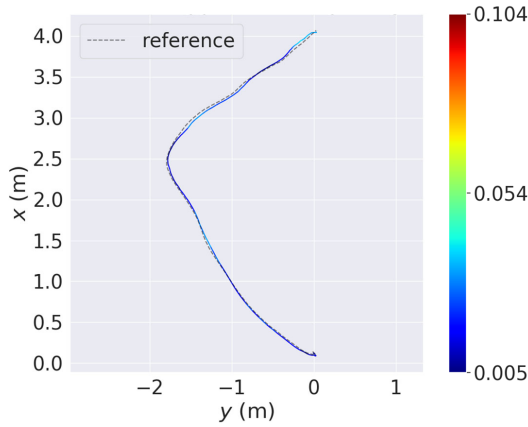


Fig. 14. Real-world experiment ground truth and odometry executed by the proposed method.

in a receding horizon fashion with the current sensor data. The planning framework can be divided into a topology-guided position trajectory generation and a perception-aware yaw angle generation. Finally, We demonstrate the superior performance of the proposed method through comparative simulation and real-world experiments. In future work, more real-world experiments will be included to verify the feasibility and novelty in different scenarios. We will also extend our perception-aware planning receding horizon method to autonomous swarms.

## REFERENCES

- [1] X. Zhang, Y. Chu, Y. Liu, X. Zhang, and Y. Zhuang, "A novel informative autonomous exploration strategy with uniform sampling for quadrotors," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 12, pp. 13 131–13 140, 2022.
- [2] X. Liu, Z. Liu, F. Kong, and F. Zhang, "Large-scale lidar consistent mapping using hierarchical lidar bundle adjustment," *IEEE Robotics and Automation Letters*, vol. 8, no. 3, pp. 1523–1530, 2023.
- [3] Y. Cao, X. Cheng, and J. Mu, "Concentrated coverage path planning algorithm of uav formation for aerial photography," *IEEE Sensors Journal*, vol. 22, no. 11, pp. 11 098–11 111, 2022.
- [4] X. Li, J. Tan, A. Liu, P. Vijayakumar, N. Kumar, and M. Alazab, "A novel uav-enabled data collection scheme for intelligent transportation system through uav speed control," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 4, pp. 2100–2110, 2021.
- [5] N. Michael, S. Shen, K. Mohta, V. Kumar, K. Nagatani, Y. Okada, S. Kiribayashi, K. Otake, K. Yoshida, K. Ohno *et al.*, "Collaborative mapping of an earthquake damaged building via ground and aerial robots," in *Field and service robotics*, pp. 33–47. Springer, 2014.
- [6] W. Zhang, J. Jia, S. Zhou, K. Guo, X. Yu, and Y. Zhang, "A safety planning and control architecture applied to a quadrotor autopilot," *IEEE Robotics and Automation Letters*, vol. 8, no. 2, pp. 680–687, 2023.
- [7] X. Zhang, J. Wang, Y. Fang, and J. Yuan, "Multilevel humanlike motion planning for mobile robots in complex indoor environments," *IEEE Transactions on Automation Science and Engineering*, vol. 16, no. 3, pp. 1244–1258, 2019.
- [8] P. Shen, X. Zhang, and Y. Fang, "Complete and time-optimal path-constrained trajectory planning with torque and velocity constraints: Theory and applications," *IEEE/ASME Transactions on Mechatronics*, vol. 23, no. 2, pp. 735–746, 2018.
- [9] J. Wen, X. Zhang, H. Gao, J. Yuan, and Y. Fang, "E3mop: Efficient motion planning based on heuristic-guided motion primitives pruning and path optimization with sparse-banded structure," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 2762–2775, 2022.
- [10] H. Wang, X. Zhang, Y. Liu, X. Zhang, and Y. Zhuang, "Svpto: Safe visibility-guided perception-aware trajectory optimization for aerial tracking," *IEEE Transactions on Industrial Electronics*, DOI 10.1109/TIE.2023.3270541, 2023.
- [11] X. Zhang, Y. Zhuang, X. Zhang, and Y. Fang, "A novel asymptotic robust tracking control strategy for rotorcraft uavs," *IEEE Transactions on Automation Science and Engineering*, DOI 10.1109/TASE.2022.3211693, 2022.
- [12] B. Zhou, J. Pan, F. Gao, and S. Shen, "Raptor: Robust and perception-aware trajectory replanning for quadrotor fast flight," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1992–2009, 2021.
- [13] J. Tordesillas and J. P. How, "Panther: Perception-aware trajectory planner in dynamic environments," *IEEE Access*, vol. 10, pp. 22 662–22 677, 2022.
- [14] J. Tordesillas and J. P. How, "Deep-panther: Learning-based perception-aware trajectory planner in dynamic environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 3, pp. 1399–1406, 2023.
- [15] V. Murali, I. Spasojevic, W. Guerra, and S. Karaman, "Perception-aware trajectory generation for aggressive quadrotor flight using differential flatness," in *2019 American Control Conference (ACC)*, pp. 3936–3943. IEEE, 2019.
- [16] S. A. Sadat, K. Chutskoff, D. Jungic, J. Wawerla, and R. Vaughan, "Feature-rich path planning for robust navigation of mavs with monoslam," in *2014 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3870–3875, 2014.
- [17] R. Wang, X. Zhang, Y. Fang, and B. Li, "Virtual-goal-guided rrt for visual servoing of mobile robots with fov constraint," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 4, pp. 2073–2083, 2022.
- [18] D. Kim, G. C. Kim, Y. Jang, and H. J. Kim, "Topology-guided path planning for reliable visual navigation of mavs," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 3117–3124, 2021.
- [19] Z. Zhang and D. Scaramuzza, "Fisher information field: an efficient and differentiable map for perception-aware planning," *arXiv preprint arXiv:2008.03324*, 2020.
- [20] R. Takemura and G. Ishigami, "Perception-aware receding horizon path planning for uavs with lidar-based slam," in *2022 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, pp. 1–8, 2022.
- [21] Z. Zhang and D. Scaramuzza, "Perception-aware receding horizon navigation for mavs," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 2534–2541, 2018.
- [22] X. Wu, S. Chen, K. Sreenath, and M. W. Mueller, "Perception-aware receding horizon trajectory planning for multicopters with visual-inertial odometry," *IEEE Access*, vol. 10, pp. 87 911–87 922, 2022.
- [23] T. Qin, P. Li, and S. Shen, "Vins-mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.
- [24] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 3529–3536, 2019.
- [25] X. Zhou, J. Zhu, H. Zhou, C. Xu, and F. Gao, "Ego-swarm: A fully autonomous and decentralized quadrotor swarm system in cluttered environments," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 4101–4107, 2021.
- [26] K. Xiao, S. Tan, G. Wang, X. An, X. Wang, and X. Wang, "Xtdrone: A customizable multi-rotor uavs simulation platform," in *2020 4th International Conference on Robotics and Automation Sciences (ICRAS)*, pp. 55–61, 2020.